**Weekly Progress Report**

**Name:** Harsh
**Domain:** Cloud Computing
**Date of submission:** June 27, 2026
**Week Ending:** 01-06

**I. Overview:** This week served as the foundational phase, focusing on understanding core cloud service models (IaaS, PaaS, SaaS), architecture, and the shift from traditional local hosting to cloud environments. My primary focus was aligning my existing Hospital Management System project with cloud scalability principles.

**II. Achievements:**

1. **Internship Familiarization:**
   - Completed modules on Cloud Computing basics, effectively distinguishing between different service and deployment models.
   - Explored cloud architecture components, specifically compute, storage, and networking essentials.

2. **Project Contributions:**
   - **Name of the project:** Hospital Management System
   - Performed a gap analysis of the current backend (SQL/Java) against cloud-native requirements.
   - Designed a preliminary database schema migration plan to support potential cloud-hosted PostgreSQL instances.

3. **Learning Progress:**
   - Acquired foundational knowledge of cloud scalability, load balancing, and the trade-offs between Virtual Machines (VMs) and containerization.

**III. Challenges:**

1. **Cloud Integration:**
    - Identifying the most cost-effective and performant cloud service model to transition the existing desktop-based Java application.
2. **Project Complexity:**
    - Resolving state-management issues when transitioning from a local SQL-connected Java application to a distributed cloud database architecture.

**IV. Learning Resources:**

1. **Internship Portal:**
    - Utilized Online Summer Internship in Cloud Computing course materials for architectural best practices.
2. **Technical Resources:**
    - Referenced AWS and Azure documentation for best practices in deploying Java-based backend systems.

**V. Next Week's Goals:**

1. **Cloud Enhancement:**
    - Begin hands-on practice by setting up a cloud sandbox environment.
    - Evaluate cloud database options (RDS or similar) to host the Hospital Management System's SQL data.
2. **Project Development:**
    - Refactor critical backend Java classes to support cloud connectivity.
    - Seek mentor feedback on the proposed cloud architecture design.

**VI. Additional Comments:** The foundational modules have provided significant clarity on why cloud-native architecture is essential for modern applications. The transition from a local desktop-bound Hospital Management System to a scalable cloud solution is a complex but vital step for my development growth.

.